

Graphical User Interface (GUI):

ETABS BUILDING AUTOMATION SYSTEM
— Advanced Structural Analysis & Design Platform —

Project **Geometry** **Stairs** **Loads** **Seismic** **Analysis** **Design** **Reports**

Building Geometry

Number of Stories:	16
Ground Floor Height (m):	4.000
Typical Story Height (m):	3.500
Number of Bays (X-Direction):	5
Number of Bays (Y-Direction):	3
Bay Spacing X (m):	6.000
Bay Spacing Y (m):	6.000

Material Properties

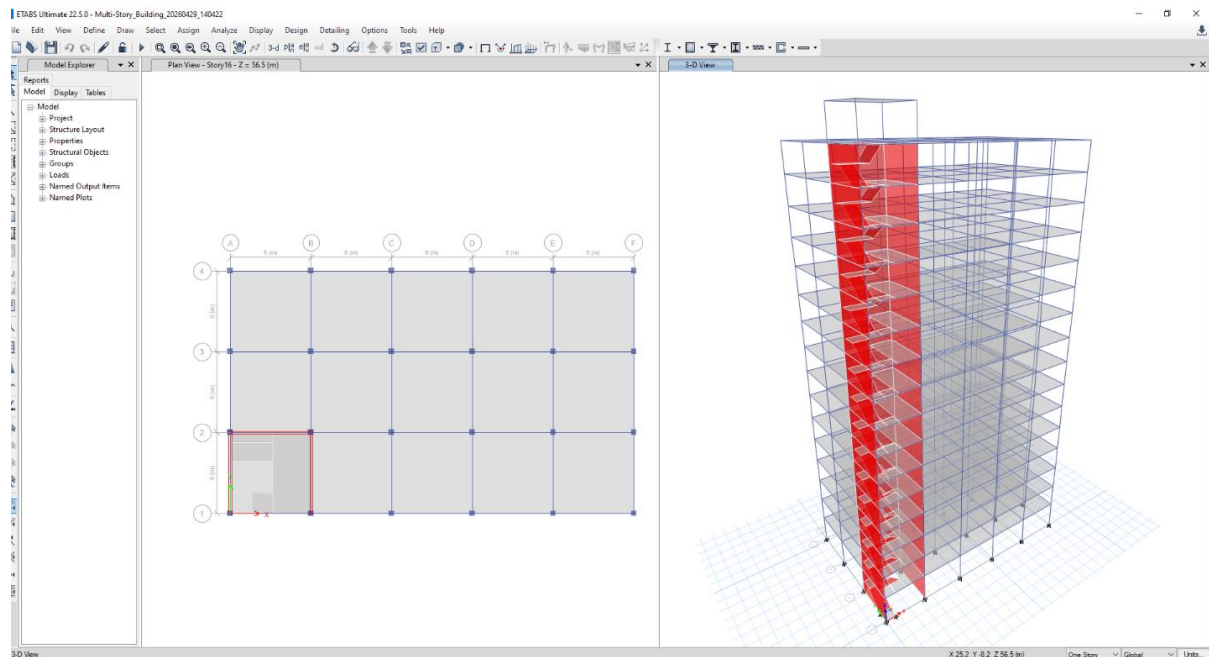
Concrete Grade f_c (MPa):	25.00
Rebar Grade f_y (MPa):	420.00

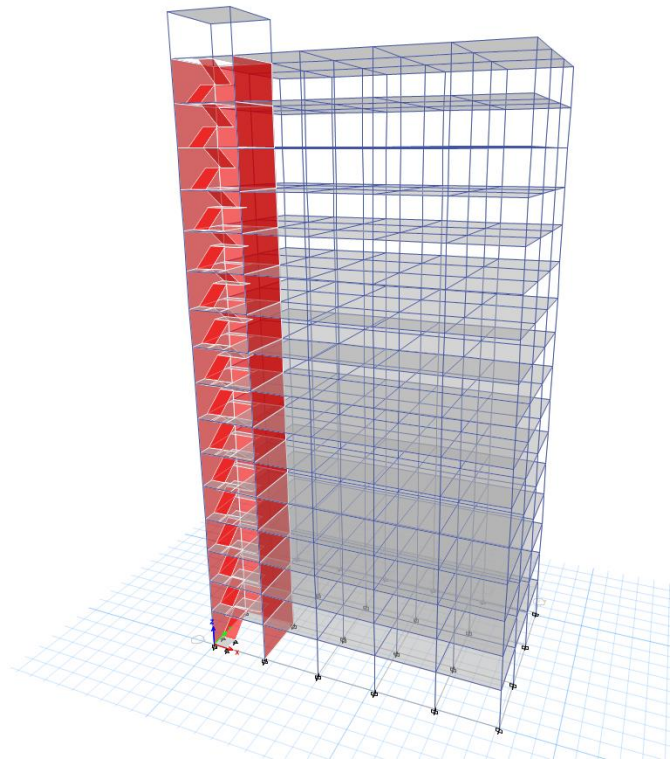
Section Properties

GENERATE ETABS MODEL **SAVE CONFIGURATION** **LOAD CONFIGURATION**

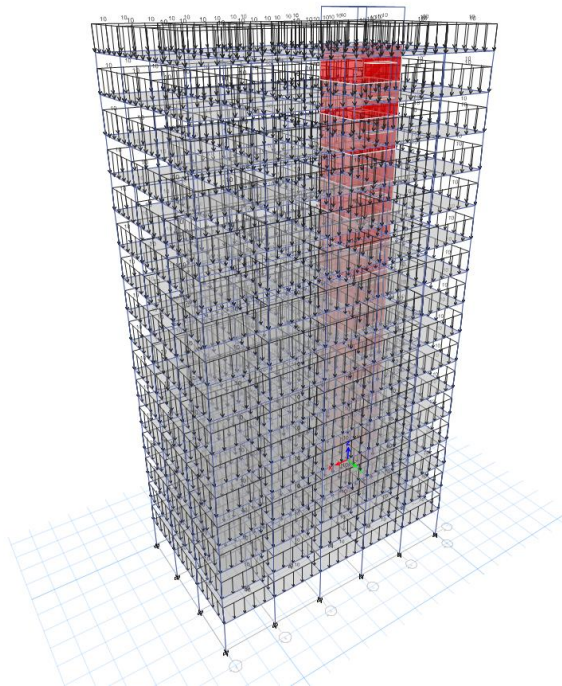
Generated 3D Structural Model in ETABS:

The structural model shown in the following figures is automatically generated using the developed graphical user interface (GUI). The user inputs essential building parameters such as the number of stories, bay spacing, material properties, and section dimensions through the application. Based on these inputs, the system utilizes the ETABS API to create the complete structural model, including beams, columns, and slabs, without manual intervention.

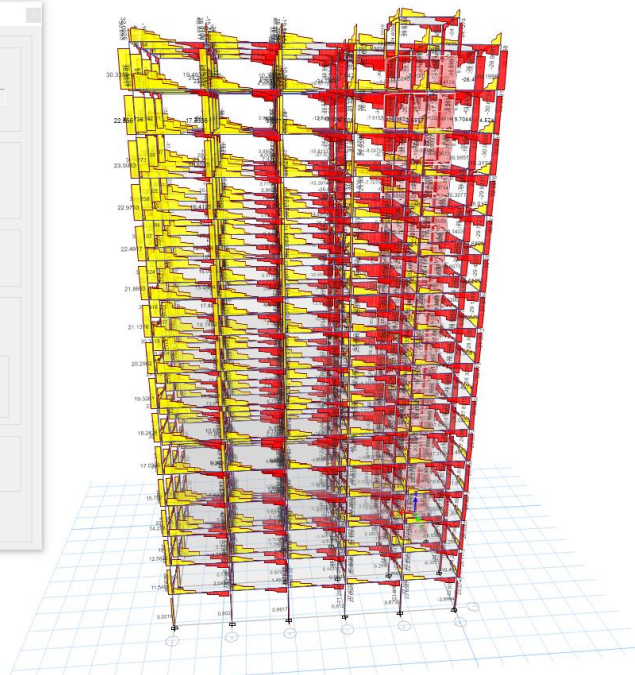
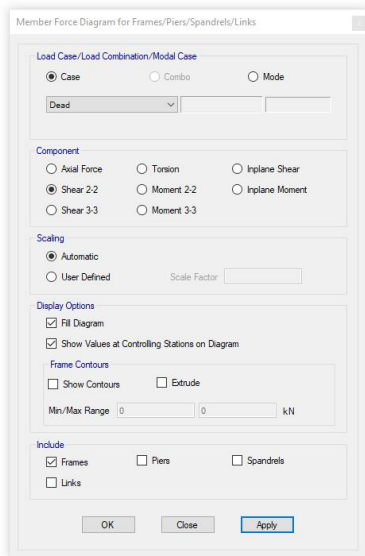
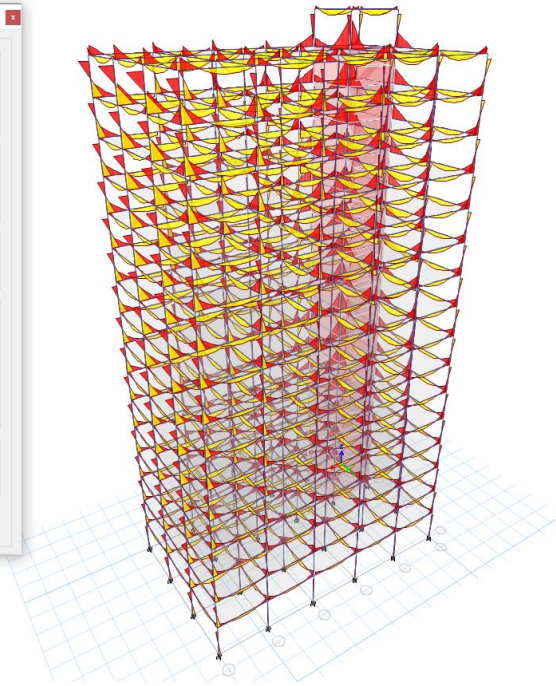
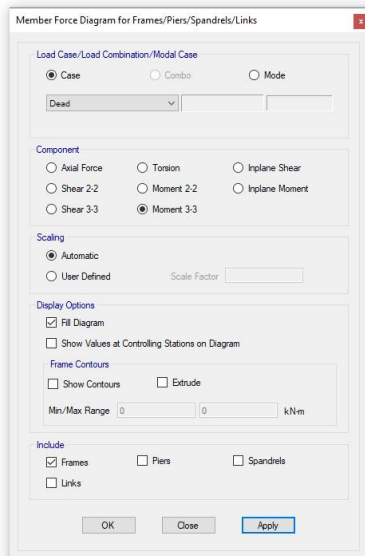




Loading Conditions in ETABS:



Analysis Results (Moments and Shear) in ETABS:



Design Stage in ETABS:

